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Born 10/01/2000

Marseille

pierreboccard.github.io

PierreBoccard

Languages

French	Native
English	C2
Croatian	Native

Computer skills

Proficient in Python, C++, Matlab, LaTeX, HTML
Machine learning: PyTorch, PyTorch Geometric (graph neural networks), Optuna
Experience with HPC, GPU

References

Dr. Alice Pisani
pisani@cppm.in2p3.fr

Dr. Nico Schuster
schuster@cppm.in2p3.fr

Dr. Frederic Courbin
frederic.courbin@epfl.ch

Dr. Anowar Shajib
ajshajib@uchicago.edu

Interests

Saxophone and piano
Photography, drawing
Tennis, sailing, basketball
Cooking

Pierre Boccard

Cosmology PhD student

Education

PhD in Cosmology — Oct 2024 - Sept 2027

CPPM, CNRS/IN2P3, Aix-Marseille Université

Thesis: *Cosmic Voids in Photometric Surveys: Systematics of the Void Size Function*. Supervisor: Dr. Alice Pisani. Funded by the ERC project COSMOBEST. Defence expected in September 2027.

Master's Thesis in Astrophysics — March 2023 - Oct 2023

University of Chicago

Measuring dark energy equation of state parameter from a compound lens system using Lenstronomy Software.

MSc. in Physics — Sept 2021 - Oct 2023

Ecole Polytechnique Fédérale de Lausanne (EPFL)

BSc. in Physics — Sept 2018 - July 2021

Ecole Polytechnique Fédérale de Lausanne (EPFL)

Collaborations

LSST Dark Energy Science Collaboration (DESC)

Lead of the cosmic voids group, which is applying to become a Topical Team

Euclid Consortium

Member

Dark Energy Spectroscopic Instrument (DESI)

Member of the void project

Publications

Toward modelling redshift uncertainties in the photometric void size function — To be submitted

The Astrophysical Journal Letters (AAS)

P. Boccard, G. Verza, G. Degni, N. Schuster, G. McFarland, K. Ghaemi Ardakani, P. Vielzeuf, A. Pisani

Cosmology from the Rubin Void Size Function — To be submitted

Astronomy & Astrophysics (A&A)

P. Boccard, G. Degni, N. Schuster, G. McFarland, K. Ghaemi Ardakani, P. Vielzeuf, A. Pisani

Measuring the dark-energy equation-of-state parameter from the compound lens system DESJ0408−5354 — To be submitted

Astronomy & Astrophysics (A&A)

P. Boccard, A. J. Shajib, F. Courbin, J. Frieman

Research visit

CCA, Flatiron Institute, New York — **July 2026**
Hosted by Dr. Francisco Villaescusa-Navarro

Talks

LSST France — **November 2025**
Les Houches Summer School — **July 2025**
LSST France — **June 2025**

Schools and Conferences

DESC Collaboration Meeting, Boston — **July 2026**
GDR CoPhy, Action Dark Energy — **2024-2026**
Les Houches Summer School — **July 2025**
Colours School, Orsay — **June 2025**
Euclid CNRS School — **July 2024**

Teaching and outreach

Public outreach, Journée des Sciences, Marseille
Full day presenting cosmology to the general public

Tutor for undergraduate students — **Sept 2019 - Jan 2023**
Ecole Polytechnique Fédérale de Lausanne (EPFL)

Mathematics and Physics teacher — **Sept 2021 - Dec 2022**
PrEP School Lausanne, private school for post-high school students

Laboratory work

Astrophysics Lab Master 3rd semester — **Sept 2022 - Feb 2023**
Research project in galaxy formation. How do supermassive black holes co-evolve with their host galaxy. The purpose of this project was to use the IllustrisTNG simulations to study host galaxy – black hole correlations

Astrophysics Lab Master 2nd semester — **Feb 2022 - July 2022**
Research project in Cosmology. Study of the clustering of voids in the BigMultiDark simulations. (grade 5.5/6)

Astrophysics Lab Master 1st semester — **Sep 2021 - Feb 2022**
Overview of research in astrophysics and cosmology through exercises. Writing of scientific article (grade 5.75/6)

Skills

Astrophysics skills
Cosmic voids and large-scale structure
Void finding with VIDE, excursion-set modelling of the void size function, Bayesian inference with nested sampling (dynesty), linear power spectra with CAMB

Simulations and mock catalogues
SkySim5000 and cosmoDC2 (Outer Rim), AbacusSummit halo lightcones, BigMultiDark, IllustrisTNG

Lenstronomy software
Advanced skills in utilizing the python software Lenstronomy for accurate and efficient analysis of strong gravitational lenses, using HST imaging

Soft skills
Writing of scientific papers, team work and management, teaching and tutoring, independence and creativity